

Technology Stack

Table 1: High-Level Technology Stack (Quick Reference)

Tech Component	Layer / Responsibility	Quality	Time to Build	Ease to Manage	Performance	Weight
React	Frontend UI framework	★★★★★	★★★★★	★★★★	★★★★	Light
Ant Design (AntD)	Unified UI component system	★★★★	★★★★★	★★★★★	★★★	Medium-Heavy
Redux Toolkit (RTK)	Global application state management	★★★★★	★★★★	★★★★	★★★★	Medium
FastAPI	Backend application framework	★★★★★	★★★★★	★★★★	★★★★★	Light
SQLAlchemy 2.0	ORM and data access layer	★★★★	★★★★	★★★★	★★★	Medium
Alembic	Database schema migrations	★★★★	★★★★	★★★★	N/A	Light
MySQL	Persistent relational database	★★★★	★★★★	★★★★★	★★★★	Medium
Redis	Cache and ephemeral runtime state	★★★★★	★★★★	★★★	★★★★★	Light
JWT Auth	Authentication and authorization	★★★★	★★★★	★★★★	★★★★	Light
Docker	Build and deployment packaging	★★★★	★★★★	★★★★	N/A	Medium
Nginx	Reverse proxy and traffic routing	★★★★	★★★★	★★★★	★★★★★	Light

Table 2: Functional Responsibility & Implementation Guidance

Tech Component	Functional Responsibility
React	Component-level local UI state, forms, modals, UI-only interactions
Ant Design (AntD)	Reusable UI components for Audience, Host, and Admin
Redux Toolkit (RTK)	Global state: session context, auth state, live poll/quiz data
FastAPI	Core backend runtime and internal orchestration
FastAPI (REST)	Auth, setup/configuration, history, analytics APIs
FastAPI (WebSocket)	Live sessions, active rooms, real-time counters
Pydantic	Request/response schemas and WebSocket payload validation
SQLAlchemy 2.0	Persistent data access and domain modeling
Alembic	Controlled database schema evolution

Tech Component	Functional Responsibility
MySQL	Durable storage for sessions, polls, votes, analytics
Redis	Ephemeral live state, counters, rate limiting
JWT Auth	Role-based access control (Admin / Host / Audience)
Docker	Consistent environments across dev, test, prod
Nginx	Routing and proxying for REST and WebSocket traffic